側邊

國立臺灣大學
資訊管理所

碩士論文 以分類強化方法協助專利維護決策之制定 林蓓妤 撰

104
2



國立臺灣大學管理學院資訊管理學研究所

碩士論文

Graduate Institute of Information Management

College of Management

National Taiwan University

Master Thesis

以分類強化方法協助專利維護決策之制定

To Renew or Abandon My Patents: An Enhanced Classification Approach for SupportingPatent Maintenance Decisions

林蓓妤

Pei-Yu Lin

指導教授：魏志平博士

Advisor: Chih-Ping Wei, Ph.D.

中華民國 104 年 2 月

February 2015

國立臺灣大學碩士學位論文
口試委員會審定書

題目：以分類強化方法協助專利維護決策之制定

To Renew or Abandon My Patents: An Enhanced Classification Approach for Supporting Patent Maintenance Decisions

本論文係林蓓妤君（學號 r01725013）在國立臺灣大學資訊管理所完成之碩士學位論文，於民國 103 年 7 月 21 日承下列考試委員審查通過及口試及格，特此證明

口試委員：吳怡瑾

林妙聆　魏志平

所　　長：李opq坤

誌謝

台大資管這個大家庭著實帶給了我一段永遠無法忘懷的歲月，而兩年的碩士生生涯怎麼一眨眼就過了；回想這一路上所受到的提攜與照顧，心中真的充滿了感謝。

本論文的完成以及碩士班的生涯中，首先要感謝的是我的指導教授魏志平老師。魏老師總是慢條斯理、有耐心的給予我們指導，指引我們一步步地解決困難並且完成研究；即便是在生活上遇到什麼挫折，魏老師也都樂於分享過來人的經驗，亦師亦友。此外，也特別感謝兩位口試委員：林怡伶老師與吳怡瑾老師，在口試時給予許多寶貴的意見，使本論文更臻完備。

此外也要感謝老師的助理虹鈞,打點我們無論是學術上或是生活上的大小事，並協助我口試順利的進行。感謝陳連進博士，在研究上給予許多建議和協助，讓我放心不少。感謝實驗室的各位好夥伴、學長姐與學弟妹，很開心能與你們共度這兩年的時光，無論開心或難過，都有你們的陪伴。感謝我的好朋友們，你們的支持與包容帶給我極大的鼓舞；當然也還要感謝諸多無法詳列的朋友們。

最後，最最最感謝的，當然是我的家人，你們始終支持我做的決定，給予我鼓勵並指引我方向，沒有你們的陪伴與幫助，我的研究生涯不會如此順利，謝謝你們。

林蓓妤 謹識

于台大資訊管理學研究所

西元二零一肆年七月

中文摘要

專利已經成為企業最重要的智慧財產以及知識財產之一，透過法律在一定期間內所賦予的合法排他權利，發明人可以保護自己的創新發明不被他人所抄襲或利用，藉以提升競爭優勢。維護費的繳納可以視為發明人獲得這項保障的方法，但是一個企業或組織內部的專利相當龐大，維護費可能造成研發與製造上的巨額負擔，因此企業傾向維護他們認為較有價值的專利，將資源更有效地分配在有前景的研發上，此時維護決策就變得相當重要且迫切。

本研究預計使用過去研究中認為會影響專利維護的因素，以資料探勘的方式建立專利維護決策支援系統，並且透過分類強化的方法，來提升判斷的準確性。這種預測及支援系統可以協助企業與組織快速的判斷是否該對專利進行維護，增進決策的效率與準確性，同時降低決策成本。此外我們也利用美國專利局資料庫中取得的專利進行實證研究，結果顯示，我們所提出的專利維護決策支援模型表現優於基準模型(未引入分類強化方法之模型)。

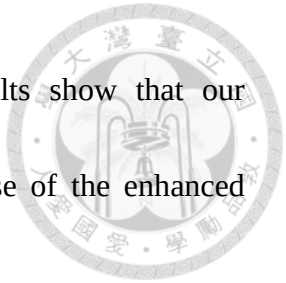
關鍵字:資料探勘、專利探勘、專利智慧、分類強化、專利維護



Abstract

Patents have become one of the most important intellectual properties and knowledge assets. With the legal exclusive rights provided by laws for a certain period, inventors can protect their inventions or products from being plagiarized or utilized, promote overall competitiveness, and gain a strategic advantage. Payments of maintenance fee can be regarded as the way inventors acquire such protection continually. However, the large portfolio of patents owned by large companies or organizations may cause heavy financial burden to them. Companies tend to maintain patents with greater value and allocate their resources to promisingtechnologies. In consequence, patent maintenance decisions are essential and urgent.

In this study, we examine factors influencing patent maintenance decisions which are considered significantby previous studies and propose a systematic solution to analyze patents for recommending patent maintenance decisions. Anenhanced classification algorithm is also applied in the study to improve the overall accuracy of our predictions. The patent maintenance decision support system can efficiently and effectivelyhelp companies and organizations determine whether a patent should be maintained while decreasing the decision making costs. In addition, we conduct experiments on the large scale United States Patent and Trademark Office (USPTO) database which contains



over millions of granted patents. The empirical evaluation results show that our proposed model outperforms the benchmark model without the use of the enhanced classification algorithm.

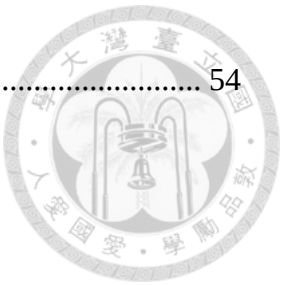
Keywords:Data Mining, PatentMining, PatentIntelligence, Enhanced Classification, Patent Maintenance,Patent Renewal

Contents

審定書 I
誌謝 II
中文摘要 III
Abstract IV
Contents VI
List of Tables IX
List of Figures X
Chapter 1 Introduction 1
1.1 Importance of patents 1
1.2 Maintenance of Patents 4
1.3 Research Motivation and Objectives 7
Chapter 2 Literature Review 12
2.1 Characteristics of Patent Maintenance 12
2.1.1 Economic Perspective 12
2.1.2 Technological Perspective 13
2.2 Related Works on Determining Patent Maintenance Decisions 14
2.3 Models 15
2.3.1 Classification 15
2.3.2 Survival Analysis 16
Chapter 3 Design of Our Patent Maintenance Prediction Technique 17
3.1 Dimensions and Variables 17
3.2 Overall Process of Our Patent Maintenance Prediction Technique 21
3.3 Learning Algorithm 22

3.4 Enhanced Classification Algorithm 24
Chapter 4 Empirical Evaluation 28
4.1 Data Collection 28
4.1.1 Target Industries and Companies 28
4.1.2 Target Patents 29
4.1.3 Maintenance Event Records 30
4.1.4 Variables Extraction 31
4.2 Evaluation Design 33
4.2.1 Benchmark Model 33
4.2.2 Evaluation Criteria 34
4.2.3 Evaluation Procedure 35
4.3 Evaluation Results 35
4.3.1 Initial Prediction Results before Enhancement 35
4.3.2 Effect of Weights on Classification Effectiveness Considering Company and IPC as Source Groups 36
4.3.3 Comparative Evaluation Results 37
4.3.4 Precision of Top and Bottom *n*% of Testing Patents 40
Chapter 5 In-depth Analysis 43
5.1 Experiment 1: Different Conditions of Company and IPC Source Groupings 43
5.2 Experiment 2: Extension of the Enhanced Model 45
Chapter 6 Conclusion and Future Directions 48
6.1 Conclusion 48
6.2 Future Directions 48
References 51

Appendix 54



List of Tables

Table 1 Current Types of Entity Status in U.S. (Source: USPTO) 5

Table 2 Current Maintenance Fees in U.S. since March 19, 2013 (Source: USPTO) 5

Table 3 Patent Renewal Rates (Source: USPTO) 7

Table 4 Maintenance Rate across Different Companies and Different IPCs 10

Table 5 Dimensions Affecting Patent Maintenance Decision by Chang (2010)..... 18

Table 6 Summary of Dimensions and Variables 19

Table 7 List of Industries in Our Research Target 29

Table 8 Profiles of Industries and Companies in Our Empirical Study 30

Table 9 Maintenance Rate across Years and Industries at Third Stage 31

Table 10 Data Sources of Variables in Our Study 33

Table 11 Formula for Accuracy, Precision, Recall, and F1 34

Table 12 Initial Prediction Results with Different Classifiers before Enhancement 35

Table 13 Prediction Results with Enhancement under Different Weights Considering Company and IPC as Source Groupings 37

Table 14 Comparative Evaluation Considering Company and IPC as Source Groups . 38

Table 15 Average Precision of Top and Bottom *n*% Patents with Year of Training/Testing Patents 1992/1993 41

Table 16 Average Precision of Top and Bottom *n*% Patents with Year of Training/Testing Patents 1993/1994 41

Table 17 Prediction Results of the Enhanced Model under Different Conditions of Company and IPC Source Groupings 44

Table 18 Number of Groups and Patents under Different Conditions 45

List of Figures

Figure 1 Utility Patents Granted per Year in U.S (Source: http://goo.gl/ZMSlfX) 2

Figure 2 Patents by Year of Grant (Source: http://goo.gl/tXn27q) 3

Figure 3 Distribution of the Most Significant Patent Fee Types in 2013 (Source: USPTO) 6

Figure 4 Overall Process of Our Patent Maintenance Prediction Technique..... 21

Figure 5 Process of Bagging Algorithm 23

Figure 6 Illustration of Changes of Probability of Maintenance after Applying the Enhanced Classification Algorithm in A Source Category..... 26

Figure 7 Distribution of Probabilities of Maintenance for Each Model 36

Figure 8 Precision and Recall of the Enhanced Model and the Benchmark Model 40

Figure 9 Comparative Evaluation under the Enhanced Model with Different α..... 47

Chapter 1 Introduction

1.1 Importance of patents

There are three categories of properties, namely chattel, real estate, and intellectual property (IP). Chattel andreal estate are tangible and have been played an important role in investment, while intellectual property is intangible and more likely to be ignored. With the evolution of civilization and technology, Intellectual Property Rights (IPR) gradually draws people's attention even though the value is sometimes difficult to estimate. Copyright, trademark, patent, and trade secret are four major types of intellectual properties.

According to World Intellectual Property Organization (WIPO), 90% to 95% invention results every year can be referenced by patents, while other technological documents can only reflect 5% to 10% of them. Patent grants monopoly rights to protect inventions or discoveries of a person or entity or for the procedures developed by such person or entity for a period of time from being plagiarized and utilized, promote overall competitiveness, and gain a strategic advantage (Shih, Liu, & Hsu, 2010; Brooks et al., 1986). In addition to the exclusive right, patents are also essential to organizations in terms of management, for instance, technological research and development portfolios, financial resource allocating and licensing, merger and acquisition decisions and

lawsuits of infringement.

As a crucial knowledge asset, more and more companies are devoted to developing patents to maintain competitiveness and gain bargaining position in litigations. The number of utility patents granted per year increases dramatically after 1983 in United State, as **Figure 1** illustrates.

Figure 1 Utility Patents Granted per Year in U.S (Source: http://goo.gl/ZMSlfX)

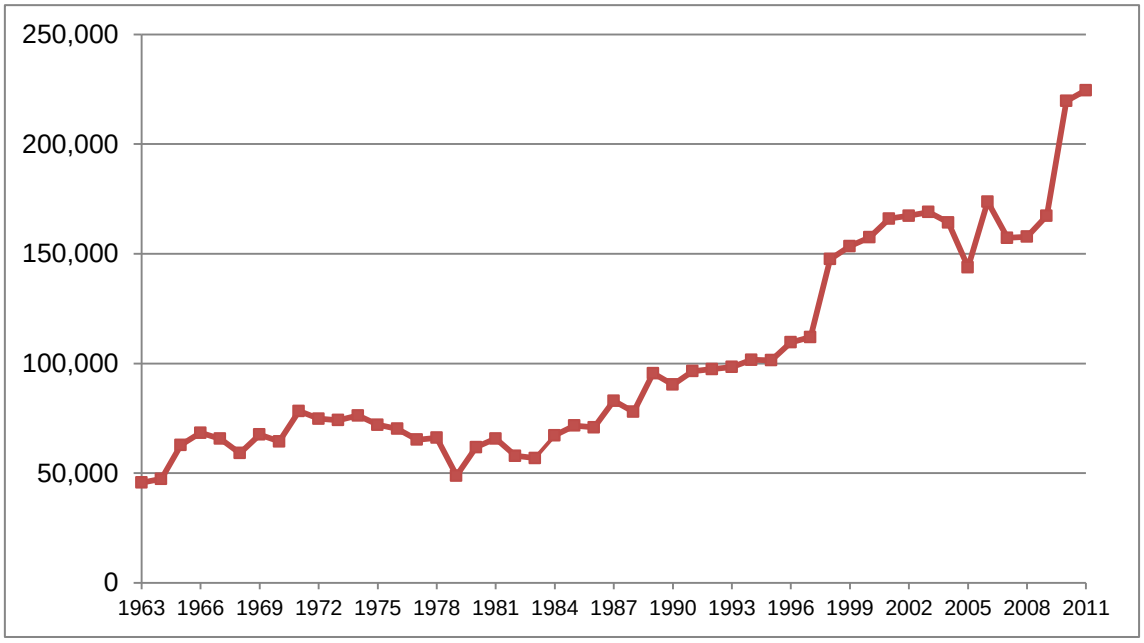


Figure 2 is a comparison chart that displays the patent growth of IBM as well as other competing technological companies such as Microsoft, Apple, General Electric, and Intel from 1998 to 2011. In 2011, IBM was granted more than 6,000 patents, while Microsoft was awarded more than 2,000 patents. General Electric and Intel were granted more than 1,000 patents, respectively, in 2011.

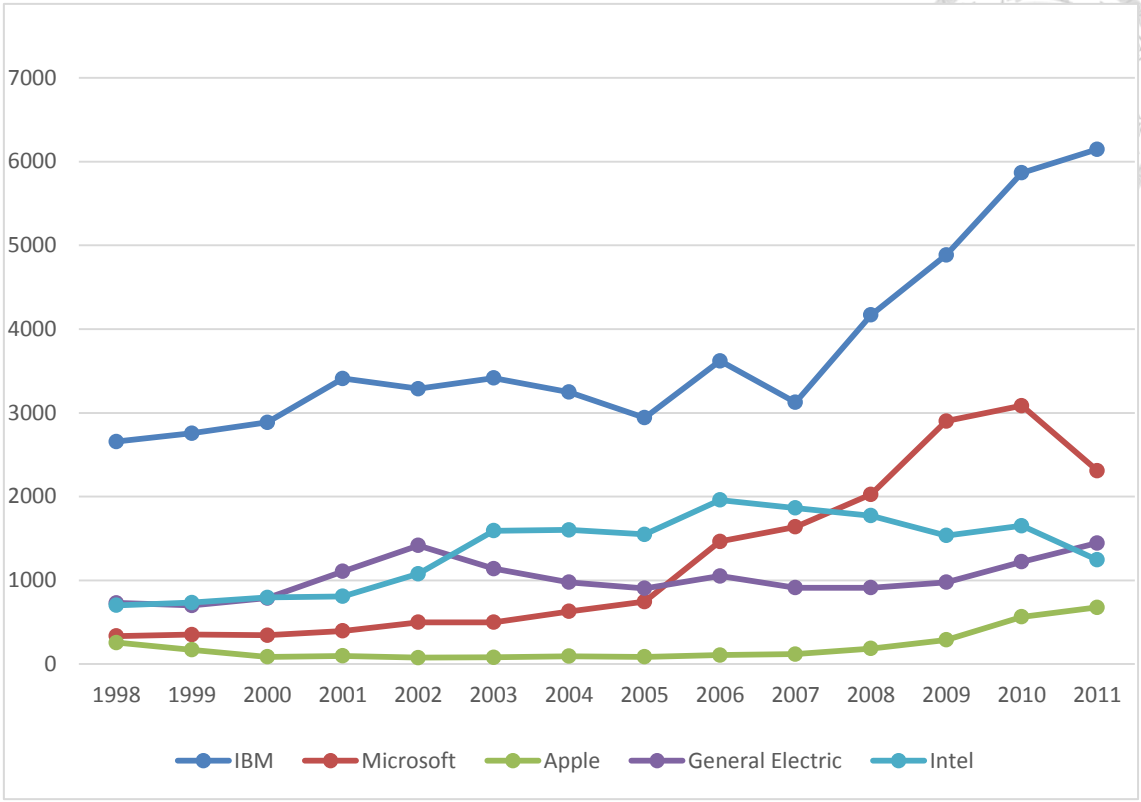


Figure 2 Patents by Year of Grant(Source: http://goo.gl/tXn27q)

In its largest acquisition ever, Google spent $12.5 billion to buy Motorola Mobility in 2011, acquiring 17,000 patents and 7,000 patent applications. Google's cache of patents is growing exponentially at a rapid rate, as MIT's Technology Review reports that they are being awarded 10 patents every day, more than Google was awarded all year when they were founded back in 2003 for comparison. Committed to having one of the world's largest patent portfolios, Google was awarded more than 1,800 patents in 2013 and became the top 10 patent recipients of the year, ahead of industrial giants like General Electric and Intel. Similarly, Hon Hai Precision Industry Co, the major assembler of Apple Inc.'s iPhones and iPads, said in the annual book that it was granted

2,279 new patents from US authorities in 2013.



1.2Maintenance of Patents

However, patent rights are not free of use forever. In the United States, the duration of protection lasts for 20 years at most from application date. In addition, patentees have to pay maintenance fees periodically in order to retain the effectual right before the expiration of the patent. A utility patent filed on or after December 12, 1980 must be renewed 3.5 (E1), 7.5 (E2) and 11.5 (E3) years after issuance of the patent. In the United Kingdom, maintenance fees are due on the fourth anniversary of the filing date and every year after that. In China, maintenance fees have to be paid every year. For a patent filed in Canada, the maintenance fees have to be paid every year subsequent to the second anniversary of the application date.

For a US patent, maintenance fee should be paid every four years with increasing amounts and depends on entity size as well. **Table 1** lists thecriteria for three types of entities since March 19, 2013. Micro entities emerge as a new entity type which are separated from small entities and are entitled to a 75% discount on most U.S Patent Office fees, while large entities have no discount. **Table 2** shows the schedule of maintenance fees in U.S. effective from March 19, 2013. For example, a micro entity has to pay $3,150 to possess full-term legal right and a large entity has to pay $12,600.

Table 1 Current Types of Entity Status in U.S. (Source: USPTO)

Entity type	Requirements	Discount rate
Micro entity	• Small Entity • Not Named on More than 4 Previously Filed Applications • Gross Income Does Not Exceed $150,162 • Not Under Obligation to Assign or License to an Entity With Income Exceeding $150,162	75%
Small entity	• Person • Any inventor or other individual that owns the patent rights individually or jointly • Small Business Concern • Any business with 500 or less employees • Nonprofit Organization	50%
Large entity	• Not qualify as a Micro Entity or a Small Entity	No discount

Table 2 Current Maintenance Fees in U.S. since March 19, 2013 (Source: USPTO)

	Due at 3.5 years	Due at 7.5 years	Due at 11.5 years	Total
Micro entity	$400	$900	$1850	$3150
Small entity	$800	$1800	$3700	$6300
Large entity	$1600	$3600	$7400	$12600

Traditionally,maintenance fees, initial application fees for filing, search, and examination, and issue fees are the major components of the revenue derived from patent operations of USPTO, which account for approximately 83% of its total patent income. **Figure 3** depicts the distribution of the most significant patent fee types of

USPTO in 2013.Patent maintenance fees are the largest source of earned revenue by fee type. During FY 2013, maintenance fees collected increased $138.7 million, or 19.9 percent, from FY 2012.

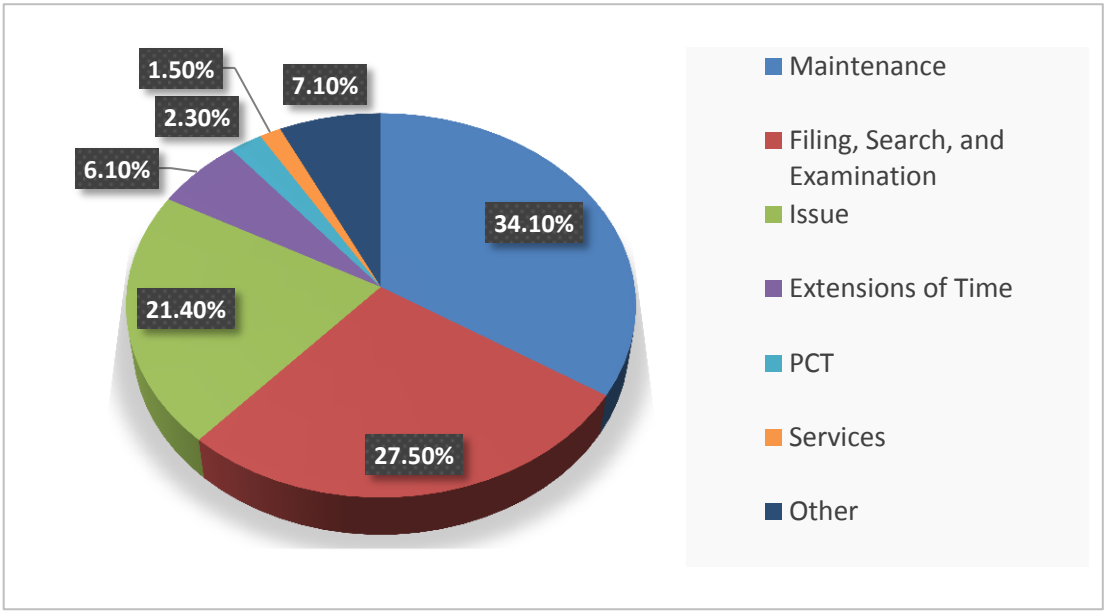


Figure 3 Distributionof the Most Significant Patent Fee Types in 2013(Source: USPTO)

Table 3 shows U.S. patent maintenance rates from 2005 to 2013. For example, in FY 2013, 92.0% of the patents issued three and half years ago were renewed, 63.3% of the patents issued seven and half years ago were renewed, and 47.0% of the patents issued 11and half years ago were renewed.

Table 3 Patent Renewal Rates (Source: USPTO)

Stage	2005	2006	2007	2008	2009	2010	2011[1]	2012	2013
First[2]	83.1%	93.1%	90.1%	83.1%	80.3%	99.4%	101.3%	78.3%	92.0%
Second[3]	65.4%	69.2%	71.4%	73.7%	63.5%	71.2%	80.6%	55.7%	63.3%
Third[4]	45.0%	44.4%	48.5%	49.2%	45.4%	50.0%	60.0%	47.0%	47.0%

1.3 Research Motivation and Objectives

High cost of renewal fees(see **Table 2**) and limited resources of firms give rise to the necessity for properly making maintenance decisions (Trappey, Hsu, Trappey, & Lin, 2006). Companies or organizations regularly examine their patents, expiring low-value patents that cannot contribute competitive advantage anymore while maintaining the ones with higher importancestrategically. This facilitates to mitigate a firm's financial burden, allow it to reallocate resources to other activities and thereby improve corporate effectiveness (Bessen, 2008). In consequence, patent maintenance decisions are essential and urgent.

1 Due to the implementation of the 15 percent fee surcharge on September 26, 2011, the FY 2011 renewal rates include some early renewals that would have otherwise been renewed in FY 2012

2 The First Stage refers to 3.5 years after issuance

3 the Second Stage refers to 7.5 year after issuance

4 the Third Stage refers to 11.5 year after issuance

Dow Chemical, for example, set up an infrastructure to organize and classify its patents to determine which ones represent strategic advantage, which presents revenue opportunities from licensing, and which patents should be abandoned to reduce the company's tax burden(Davenport, DeLong & Beers, 1998). By managing its patents more effectively, Dow Chemical saved $4 million in its first year of a review program, and expected to generate more than $100 million in licensing revenues that might otherwise have been forgone. Companies such as Microsoft, Texas Instruments, and IBM (with 3,000 percent increase in revenues from licensing its intellectual capital) have also demonstrated the power of licensing to improve company's financial performance(Jones, Norris, & Solomon, 2002).

Another reason that companies should properlymake maintenance decisions is to gain protection from patent right or bargaining power from litigations and negotiations as a strategic tool. Patents help ensure that the payoff from R&D and the patent owner's competitive advantage are maximized(Choksi, 1999).

There is no universal guideline to maintenance decisions. In addition, making the decisionsusually consumes lots of time and resources.There are plentiful of researches probe into the factors that would influence patent maintenance decisions. According to



Jin et al., different owners have different strategies for patent maintenance in terms of patent quality, novelty, technique value, technique relevancy, and economic value(Jin et al., 2011). Chang conducted aDelphi study to investigate the dimensions and factors affecting patent maintenance decisions(Chang, 2010).

The potential factors mentioned above are merely some of the overall consideration. Moreover, previous research studies did not consider the implicit similarity information existing among patents to be predicted. As**Table 4** shows, maintenance rates of patents issued in 1993 among different companies and 3-digit[5] IPC classifications vary apparently. We conjecture that the maintenance decisions are highly related to some implicit similarity such as companies, IPCs, or industries.

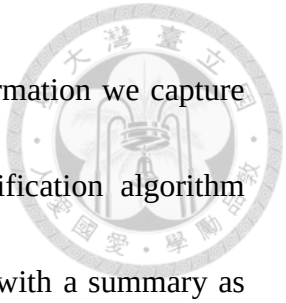
[5]IPC classification code is a hierarchical system consisting of section, subsection, class, subclass, group, and subgroup. 4-digit IPC stands for section, class, and subclass (ex. A63H) while 3-digit IPC means section, class (ex. A63)

Table 4 Maintenance Rate across Different Companies and Different IPCs

Company/IPC	B05	G06	H01
Hitachi	42.86%	42.66%	49.28%
Toshiba	75.00%	49.48%	71.75%
Siemens	22.22%	51.52%	34.62%
Sony	80.00%	75.47%	60.34%

In consequence, we draw the concept of integrating catalogs proposed by Agrawal andSrikant(2011) and apply the enhanced classificationalgorithm to our method.In this study, we expect to examine indicators possibly influencing patent maintenance decisions and propose a data-mining based technique for patent maintenance prediction. With the enhanced classificationalgorithm to improve overall accuracy of prediction, our proposedtechnique can assist companies and organizations in promoting the effectiveness and efficiency of patent maintenancedecision making and reducing decision costs.

This chapter discusses the research background, motivation and objectives of this thesis study. The rest of the thesis is organized as follows. Chapter 2 reviews literature related to this study. In Chapter 3, we depict the proposed data-mining based patent maintenance prediction technique. Chapter 4 shows our empirical evaluation, including data collection and comparative evaluation results. In Chapter 5, we conduct in-depth



analysis to further investigate how different implicit similarity information we capture and how different conditions when applying the enhanced classification algorithm would influence prediction results. Finally, we conclude this study with a summary as well as some future directionsin Chapter 6.

Chapter 2 Literature Review

In this chapter, we will introduce how firms and experts evaluate maintenance decision in both economic and technological perspectives. Some factors are proved to be highly relevant to patent maintenance decisions in prior studies which provide essential insight and criteria in determining whether patents should be maintained or not. Following are some related works on determining maintenance decisions. Besides, we will compare two streams of patent maintenance research, i.e.,classification and survival analysis.

2.1 Characteristics of Patent Maintenance

Most of the studies on patent maintenance tend to evaluate decisions in terms of technology perspective and economy perspective. The primary concern of economics is the commercial value of such innovations to companies and organizations, while technological analysts seek the opportunities to identify innovations which facilitate important technological advances.

2.1.1 Economic Perspective

The maintenance decision of firms to renew their patents is largely assumed to be the result of a careful valuation exercise to balance their expected revenues with renewal fees(vanZeebroeck, 2009). Thus, it opens up an opportunity for economics and

researchers to estimate the potential value of patents. However,most patents are of low value, with only a few ones having a high value on the contrary(Pakes, 1986). Grilichesestimates that the contribution of the number of patents of a firm to the variance in the value of the firm is onlyon the order of 7%(Griliches, 1998). Econometric studies have confirmed that renewal fees influence the maintenancedecision and that more valuable patents are held longer(Schankerman & Pakes, 1985).Bessenused renewal data to estimate the economic value of U.S. patents and found out that patents issued to large corporations are more valuable than those issued to small patentees. Also, litigated patents are more likely to be maintained(Bessen, 2008).

2.1.2 Technological Perspective

Patents with significant technological impacts will tend to have an influence upon large numbers of later patents, and will therefore be cited regularly. Patent citation counts have been one of the most important and widely used indicators for estimating patent value and technological importance (Bessen, 2008; Harhoff, Narin, Scherer, & Vopel, 1999). Thomassuggested that a significant positive relationship between patent citations and patent renewals has existed ever since maintenance fees were introduced in the United State in 1980(Thomas, 1999). Albert et al.foundasignificant relationship

between the assessments of technological impact and the number of citations received by direct surveys of patent owners at Eastman Kodak(Albert & Avery, 1990). Harhoffet al.showed that patent citations are correlated with patent value obtained from a survey of German patent holders and further effect patent maintenance decisions(Harhoff et al., 1999).

To determine whether the patent is worth maintaining, companies sometimes have to take the whole patent portfolio into consideration instead of focusing on single patents. Patent holders are paying more attentions to the strength and value of their overall portfolio, increasingly used to negotiate cross-licenses, rather than to the quality and value of each individual filing.Liu et al. useddivisional and continuation patent applicationsto findthe interrelation between patents, and confirmed that interrelated patents are more valuable than others stand-alone ones by examining the dataset of US pharmaceutical and biotechnology patents; therefore more likely be renewed(Liu, Arthurs, Cullen, & Alexander, 2008).

2.2 Related Works on Determining Patent Maintenance Decisions

Previous studies exploreda variety of indicators which influence renewal decisions. The research byvan Zeebroeck showed that the classical patent value proxies such as families, claims, IPC classes and forward citations, constitute strong predictors of the

entire length of patent rights(van Zeebroeck, 2009).Oh et al.differentiated patent citations along two dimensions, namely assignees and technologies, into four types, and propose a weighted citation approach for assessing and ranking patents(Oh, Lei, Mitra, & Yen, 2012). Yoon and Parkconsidered using patent citations to represent the interrelated relationship of patents, as a result, they proposed a network-based patent analysis as an alternative method(Yoon & Park, 2004). Similarly, Wang et al.used the social network approach to measure patent technology flow instead of perceiving through citing and cited of patents since the knowledge flow among patents has not been considered(Wang, Lo, Liao, & Lin, 2012).

2.3Models

2.3.1 Classification

Numerous classification tasks based on patent content have been conducted in order to help assign patent applications to the right examiners or classify patents to patent classification schemes(Trappey et al., 2006; Fall et al., 2003; Li, Chen, Zhang, Li, & Nunamaker, 2009). We can formulate patent maintenance decisions as a classification problem considering whether to maintain or abandon a patent. Jin et al.built a classification model with network-based refinement approach to automatically recommend patent maintenance decisions(Jin et al., 2011) while Changapplied theopen

source data mining software Weka to build a classifier to generate maintenance predictions(Chang, 2010).

2.3.2 Survival Analysis

Apart from classification models, some researchers made use of survival analysis to examine the survival time of patents since it can analyze event durations and thus is suitable for evaluating time-series data like patent life time. The main concept of survival analysis is the risk portfolio. Moreover, a survival model is a good approach to deal with the censored data. Wang et al.incorporated the social network analysis to determine patent value and conducted survival analysis to investigate the probability for a patent is renewed(Wang et al., 2012).Jun & Uhm (2013)proposed a predictive model of patent registration time using survival analysis. Nikzad studied the lifetime and grant time of patents in Canada andsuggested that non-PCT patents and more complex patents survive longer and also need more time to be granted(Nikzad, 2011). Also, there is a statistically significant difference among IPC main classes in terms of survival time.van Zeebroeck employed patent applications filed to the EPO over a 20 years period and analyzed with a survival model the determinants of their maintenance throughout their life, from filing to withdrawal, refusal or lapse(van Zeebroeck, 2009).



Chapter 3 Design of Our Patent Maintenance Prediction Technique

We consider patent maintenance as a classfication problem and employ a data mining method to develop our patent maintenance prediction technique. In the following, we first describe the variables we adopt for patent maintenance prediction and then depict the detaileddesign of our patent maintenance prediction technique.

3.1 Dimensions and Variables

The fisrt step of constructing the model is to identify suitable and representative variables or factors that possibly influence patent maintenance decisions. On the basis of Chang'sresearch (2010) (see **Table 5**), which conducteda Delphi study to identify influential dimensions of patent maintenance decision from a group of experts in the related fields, we match related variables such as citations count and family size from the literature review to those dimensions. Afterwards, we learn a patent maintenance classifacation model froma given set of training instances.

Table 5 Dimensions Affecting Patent Maintenance Decision by Chang (2010)

Dimension	Brief Description
Business Strategy	Direction of the company's business, marketing strategies, and future competitiveness
Patent Use	Use for production, and probability of future use
Commercialization	Whether it is utilized or embedded in a final product, and the possibility of the patented technology being commercialized
Market Scale	Scale of the market
Technology Transfer & Licensing	Possibility of patent licensing, and probability of future transfer
Patent Family	Whether the patent belongs to a core patent family, whether some other patents are extended from the patent, and whether the patent is applied in multiple countries
Patent Quality	Whether the patent is influenced by prior arts
Patent Defensiveness	Scope of the claims, difficulty of design around, possibility of preventing litigations from patent troll.
Technology Type	Technology lifecycle of the patent, whether the patent belongs to part of the technology standard, how advance the technology protected by the patent is, and whether the technology protected by the patent is a core technology
Future R&D	Possibility of future innovation extended from the patent
Patent Age	Remaining life of the patent

Table 6liststhedimensions and corresponding variables adopted in our research.In addition, reference sources of these variables are shown in **Exhibit 1**. There are 23 factors grouped into 7 dimensions including*Business strategy, Commercialization, Patent Family, Patent Defensiveness, Technology Type, Future R&D, and Industry*.

Table 6 Summary of Dimensions and Variables

Business Strategy	
Maintenance compatibility of tech scope (IPC)	$\frac{\sum_{i=1}^{m} \frac{\vec{X} \cdot \overrightarrow{M_i}}{\lvert\vec{X}\rvert \lvert\overrightarrow{M_i}\rvert}}{m}$ $\vec{X}$ = the vector of IPC classification codes of the focalpatent $\overrightarrow{M_i}$ = the vector of IPC classification codes of maintained patent i with the same assignee last year m = the number of maintained patents with the same assignee last year
Abandonment compatibility of tech scope (IPC)	$\frac{\sum_{i=1}^{a} \frac{\vec{X} \cdot \overrightarrow{A_i}}{\lvert\vec{X}\rvert \lvert\overrightarrow{A_i}\rvert}}{a}$ $\vec{X}$ = the vector of IPC classification codes of the focal patent $\overrightarrow{A_i}$ = the vector of IPC classification codes of abandoned patent i with the same assignee last year a = the number of maintained patents with the same assignee last year
Commercialization	
Patent scope(IPC)	# of different four-digit IPC classification codes of the focal patent
Patent scope (USPC)	# of different three-digit USPC classification codes of the focal patent
Patent Family	
Family size (countries)	# of unique countries that are in the family
Family size (applications)	# of applications the patent protecting a particular invention
Major market	# of applications in three patent offices/3 (USPTO/EPO/JPO)
Number of backward citations	# of patents cited by the focal patent
Number of backward citations (US patents)	# of US patents cited by the focal patent
Science links	# of other citations cited by the focal patent

Number of forward citations within 5 years	# of patents citing the focal patent within 5 year after filing date
Patent Defensiveness	
Number of claims	# of claims of the focalpatent
Number of independent claims	# of independent claims of the focalpatent
Approval time	= grant date – application date
Technology Type	
Originality index (by IPC)	$1 - \sum_{c} \left(\frac{\#\text{ of backward citations with IPC c}}{\#\text{ of backward citations}} \right)^2$
Generality index (by IPC)	$1 - \sum_{c} \left(\frac{\#\text{ of forward citations within 5 years with IPC c}}{\#\text{ of forward citations within 5 years}} \right)^2$
Cited half-life (by median)	the median patent age of backward citations
Maintenance of backward citations	$\frac{0 \times \sum E_1 + 1 \times \sum E_2 + 2 \times \sum E_3 + 3 \times \sum M}{\text{the number of backward citations}}$ E_i = # of patents cited by the focal patent that are expired at ith stage M = # of patentscited by the focal patent that aremaintained at all stage
Number of cross state cooperation	# of different states (countries) in which theinventor(s) of the focal patent resides
Future R&D	
Internal sequential innovation	"yes" if the patent belongs to sequence innovation (division, continuation, CIP) or "no" otherwise
Forward self-citation ratewithin 5 years	$\frac{\#\text{ of forward citations with the same assignee within 5 year after filing date}}{\#\text{ of forward citations within 5 year after filing date}}$
Backward self-citation rate	$\frac{\#\text{ of backward citations with the same assignee}}{\#\text{ of backward citations}}$
Industry	
Industry	Nominal values for different industries in our data set

3.2 Overall Process of Our Patent Maintenance Prediction Technique

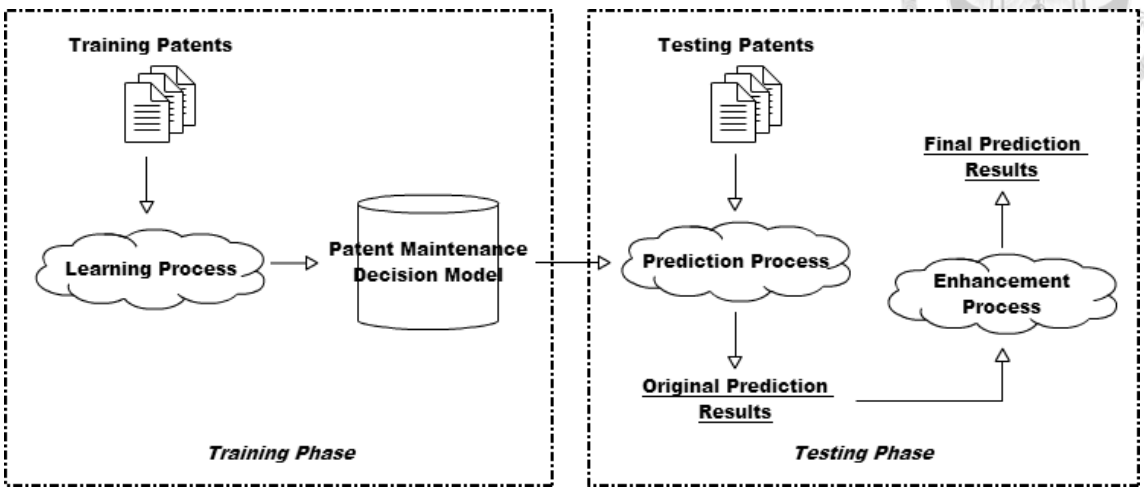


Figure 4 Overall Process of Our Patent Maintenance Prediction Technique

Classification involves in utilizing a set of features or parameters to characterize each patent using a supervised learning algorithm. Supervised learning means that a human expert has determined into what classes an object be categorized and also has provided a set of sample objects with known classes. Asupervised learningalgorithmthen uses this set of known objects called the training set to learn how to classify objects into different classes.Decision tree, NaïveBayes classifier, backpropagation neural network, *k*-nearest neighbor classifier, and support vector machine (SVM) are popularsupervised learningalgorithms.

Our proposed patent maintenance prediction technique consists of two main phases (please see **Figure 4**). In the training phase, the training set is used to induce a classification model (referred to as the patent maintenance decision model) for

separating the two classes of patent maintenance, namely ‘maintain’ (or renew)and ‘abandon’ (not maintain). In the testing phase, the patent maintenance decision modelinduced in the previous phase is applied to a set of patents that do not have known decisions yet.

We use patents granted in 1992 as training data to test patents granted in 1993. Similarly, patents granted in 1993 were utilized to train a patent maintenance decision model to recommend maintenance decisions of patent granted in 1994.

After the initial prediction results, we then apply anenhanced classification algorithm(Agrawal & Srikant, 2001) to improve the performance of the classification as our final prediction results.

3.3 Learning Algorithm

We apply open source data mining software Weka to construct a classifier with Naïve Bayes as the underlying learning algorithm. In order to improve prediction effectiveness, we apply Bootstrap Aggregating, known as the bagging algorithm, which was proposed by Breiman in 1996. Bagging leads to improvements for unstable procedures and give substantial gains in accuracy(Breiman, 1996). It randomly generates*k*subsets of cardinality from the original training data set with replacement (an element may appear

multipletimes in subsets) and build a classifier through the same underlying learning algorithm (Naïve Bayes in our study) for each subset. The prediction results derived from individual subsets are then combined by the voting method to arrive at an overallprediction(**Figure 5**). To avoid a tie, we choose an odd number k = 15.

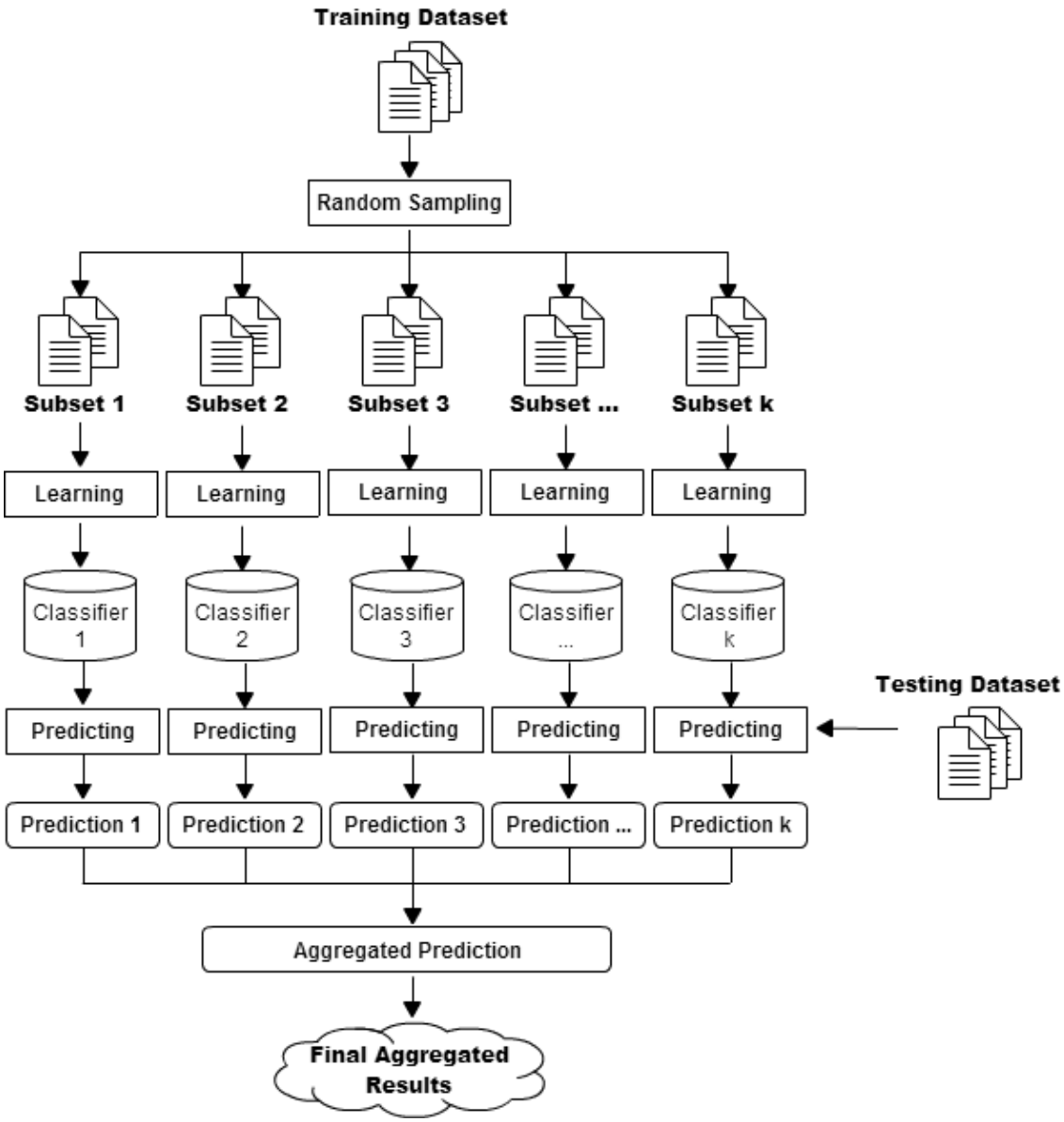


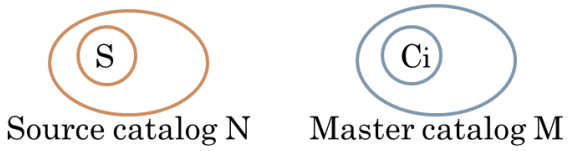
Figure 5 Process of Bagging Algorithm

3.4 Enhanced Classification Algorithm

Addressing the problem of integrating documents from a source catalog(which consists of multiple source categories) into a master catalog (which consists of multiple master categories), Agrawal and Srikant(2001) showed how Naïve Bayes classification can be boosted by incorporating the implicit similarity information presented in the source categories. The intuition is that if two documents belong to the same category in the source catalog, they are more likely to belong to the same category in the master catalog.

The original Naïve Bayes classifier estimates the posterior probability of the master category C_i given a document *d* via Bayes' rule

$$\mathrm{P}(C_i|d) = \frac{\mathrm{P}(C_i)\mathrm{P}(d|C_i)}{\mathrm{P}(d)} \qquad (Eq.1)$$



Let *S* denote a source category in the source catalog *N*, andC_i denote a master category in the master catalog *M*. The posterior probability of the master category C_i in *M* given a document *d*and the fact that *d*belongs to the source category *S* in *N* is computed as

$$\mathrm{P}(C_i|d,S) = \frac{\mathrm{P}(C_i|\mathrm{S})\mathrm{P}(d|C_i)}{\mathrm{P}(d|\mathrm{S})} (Eq.2)$$

$$assuming\ d, S\ are\ independent\ given\ C_i$$

Equation 1 and **Equation 2** are similar except that we have $\mathrm{P}(C_i|\mathrm{S})$in Equation 2 instead of $\mathrm{P}(C_i)$ in Equation 1 (we also have $\mathrm{P}(d|\mathrm{S})$ in Equation 2 instead of $\mathrm{P}(d)$ in Equation 1 in the denominator, but since the term is the same for all classes, it does not affect relative probabilities). To estimate $\mathrm{P}(C_i|S)$, we first classify the documents using the original Naïve Bayes algorithm,then use the categories of the documents in S to compute the estimate in **Equation 3**. Positive index *w* is the weight to give to the implicit information that reflects the similarity between the categorization of the two catalogs.

$$\mathrm{P}(C_i|S) = \frac{|C_i| \times (\#\ of\ docs\ in\ S\ predicted\ to\ be\ in\ C_i)^w}{\sum_{j=1}^{n}(|C_j| \times (\#\ of\ docs\ in\ S\ predicted\ to\ be\ in\ C_j)^w)} (Eq.3)$$

We would like $\mathrm{P}(C_i|S)$ to be larger for the category with more documents while smaller for the category with fewer documents. When w = 0, the prediction of the enhanced classifier will be the same as the prediction by the original classifier.

It is proved that for each document *d*, there either exists a single interval(W_1, W_2), $0 \le W_1 \le W_2$ in which the document will be correctly classified or never be correctly classified(Agrawal & Srikant, 2001). It shows the superiority of the enhanced classification algorithm.

In our study, S is designed to be one of the source categories among testing patentssuch as whether they belong to the same assignee and the same IPC classification simultaneously. C_i isone of the two classes of patent maintenance: 'maintain' or 'abandon.'**Figure 6**illustrates the changes of the probability of maintenance of each patent in a source category after applying the enhanced classification algorithm. We conjecture that the distribution of maintenance probabilities of all testing patents in a source categorywill move in accordance with the majority of the initial predictions in the same source category.

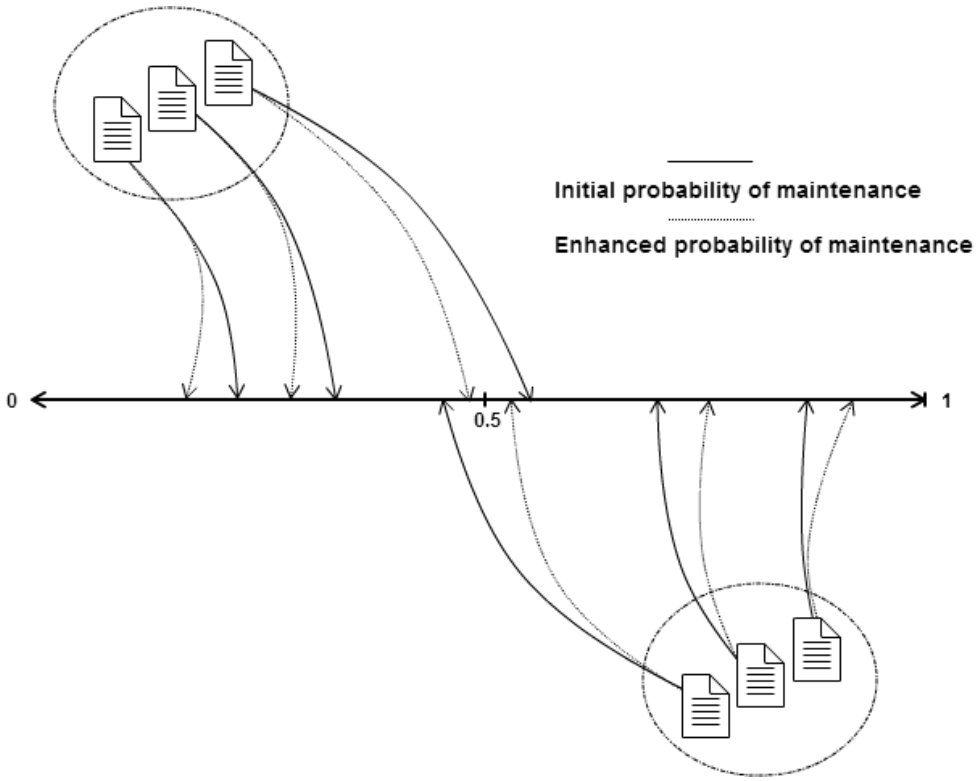


Figure 6 Illustration of Changes of Probability of Maintenance after Applying the Enhanced Classification Algorithm in A Source Category

After we obtain the initial prediction for each testing patent as well as its predicted probabilities of maintenance and abandonment, we then multiply the corresponding $P(C_i|S)$ to the initial probabilities respectively. After normalizing the new distribution of probabilities, we can achieve the final prediction result according to the new probabilities.For those patents that belong to more than one source group, we average the probabilities retrieved from the multiple enhancement processes.

Chapter 4 Empirical Evaluation

4.1 Data Collection

4.1.1 Target Industries and Companies

First, we have to decide our research target. Based on Chang's(2010) study, we first select industries and corresponding companies to be included in our research. According to Globe 500[6] in 2005, there are 50 industry sectors in total. However, some industries such as Banks and Insurance have few patents. As a result, these industries are removed from our research target. Furthermore, we exclude those industries with less than 3 companies listed in Global 500as well since they are not so representative. Thus, we have 8 industries (see **Table 7**) in our empirical evaluation including Aerospace and Defense, Chemicals, Electronics, Electrical Equipment, Motor Vehicles & Parts, Network and Other Communications Equipment, Pharmaceuticals, Semiconductors and Other Electronic Components, and Telecommunications. Among these industries, there are 69 companies listed (See **Exhibit 2**).

[6] http://fortune.com/global500/2005/wal-mart-stores-inc-1/

Table 7 List of Industries in Our Research Target

Industry	# of companies
Aerospace and Defense	10
Chemicals	9
Electronics, Electrical Equipment	10
Motor Vehicles & Parts	10
Network and Other Communications Equipment	5
Pharmaceuticals	10
Semiconductors and Other Electronic Components	5
Telecommunications	10
Total	**69**

4.1.2 Target Patents

Second, we search patents belonging to our target companies.Because the number of patents increases dramatically after 2000 and we apply a 5-year time interval in some of our variables computation (e.g., forward citations), we focus on those patents that were granted in 1992, 1993, and 1994 from USPTO website.These patents faced the third stage maintenance decisions in 2003 to 2006.Besides, we ignorethe patents of a companyfor a particular year if the number of patents of the company was less than 10 at that year. The remaining patents should have encountered the third stage of maintenance decisions between 2003 and 2006 if they were maintained in the first and the second stage. **Table 8** shows the profiles of patents in our empirical experiment with

a total of 21,908 patents.

Table 8 Profiles of Industries and Companies in Our Empirical Study

Industry	1992	1993	1994	Total
Aerospace and Defense	361	309	290	960
Chemicals	672	720	654	2,046
Electronics, Electrical Equipment	3117	3604	3852	10,573
Motor Vehicles & Parts	692	673	537	1,902
Network and Other Communications Equipment	675	703	790	2,168
Pharmaceuticals	292	358	366	1,016
Semiconductors and Other Electronic Components	461	507	690	1,658
Telecommunications	455	480	650	1,585
Total	**6,725**	**7,354**	**7,829**	**21,908**

4.1.3 Maintenance Event Records

We obtain the maintenance records of each patent in our dataset from USPTO website[7]. Retrieved data contains patent information such as patent number, application number, entity status, filing date, issue date as well as event entry date, and event codes. Among hundreds of events codes, we merely focus on the third-stage maintenance decision.

[7] https://eipweb.uspto.gov/MaintFeeEvents/

"Payment of maintenance fee, 12th year" means that the patent was maintained in 3rd stage while "Patent expired for failure to pay maintenance fees" means the patent was abandoned. In **Table 9**, themaintenance rate of across industries and years are provided.

Table 9 Maintenance Rate across Years and Industries at Third Stage

Industry	# of Companies	1992	1993	1994	Average
Aerospace and Defense	10	0.83	0.79	0.81	0.81
Chemicals	9	0.60	0.60	0.62	0.61
Electronics, Electrical Equipment	10	0.76	0.66	0.71	0.71
Motor Vehicles & Parts	10	0.62	0.58	0.61	0.60
Network and Other Communications Equipment	5	0.71	0.68	0.71	0.70
Pharmaceuticals	10	0.69	0.63	0.56	0.62
Semiconductors and Other Electronic Components	5	0.91	0.92	0.94	0.93
Telecommunications	10	0.81	0.87	0.92	0.87
Total/Average	**69**	**0.74**	**0.68**	**0.73**	**0.72**

4.1.4 Variables Extraction

The variables involved in our patent maintenance prediction technique can be derived from multiple online websites and tools. In addition to USPTO website, NBER

website[8](Hall et al., 2001), Family-izer™ website[9], and Globe 500 contains numerous patent characteristics which can be utilized to compute the value of variables.**Table10** lists the sources of each variable.

USPTO provides detailed patent information such as inventors, assignees, claims, citations, and full-text patents of United State. NBER comprises detailed information on almost 3 million U.S. patents granted between January 1963 and December 1999, all citations made to these patents between 1975 and 1999 (over 16 million), and a reasonably broad match of patents to Compustat (the data set of all firms traded in the U.S. stock market). Finally, Family-izer™website is a patent family retrieval tool containing the total number of family members and the number of unique countries and/or regionsthat are in the family. Globe 500 is an annual ranking of the top 500 corporations worldwide as measured by revenue. The list is compiled and published annually by *Fortune* magazine.

[8]http://www.nber.org/patents/
[9]http://www.familyizer.com/

Table 10 Data Sources of Variables in Our Study

Dimensions	Variables	Sources
Business strategy	Maintenance compatibility of tech scope (IPC)	USPTO
	Abandonment compatibility of tech scope (IPC)	USPTO
Commercialization	Tech scope (IPC)	USPTO
	Tech scope (USPC)	USPTO
Future R&D	Internal sequential innovation	USPTO
	Forward self-citation rate	NBER / USPTO
	Backward self-citation rate	NBER / USPTO
Patent Defensiveness	Number of claims	USPTO
	Number ofindependent claims	USPTO
	Approval time	USPTO
Patent Family	Number ofbackward citations_US	USPTO
	Number ofbackward citations	USPTO
	Science links	USPTO
	number offorward citations	NBER / USPTO
	Family size (countries)	Family-izer™
	Family size(applications)	Family-izer™
	Major market index	Family-izer™
Technology type	Generality index (IPC)	NBER / USPTO
	Originality index (IPC)	NBER / USPTO
	Cited half-life (median)	USPTO
	Maintenance of backward citations	USPTO
	# of cross state cooperation	USPTO
Industry	Industry	Globe 500

4.2 Evaluation Design

4.2.1 Benchmark Model

To compare with the use of the enhanced classification algorithm, we designate the

original classification algorithm (i.e., without the use of the enhanced classification algorithm) as our benchmark model in the following experiments.

4.2.2 Evaluation Criteria

Accuracy, precision, recall, and F1 measurements are four common indicators to evaluate the predicted outcomes. Each testing instance can be categorized into one of the followingconditions, namely true positive, true negative, false positive, and false negative according to their ground truth and predicted outcome. The four indicators are computed as **Table 11**.

		Ground Truth	
		yes	no
Predicted Outcome	yes	True Positive TP	False Positive FP
	no	False Negative FN	True Negative TN

Table 11 Formula for Accuracy, Precision, Recall, and F1

Indicators	**Formula**
Accuracy	$\frac{(\mathrm{TP}+\mathrm{TN})}{(\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN})}$
Precision	$\frac{\mathrm{TP}}{(\mathrm{TP}+\mathrm{FP})}$
Recall	$\frac{\mathrm{TP}}{(\mathrm{TP}+\mathrm{FN})}$
F1	$\frac{2 * \mathrm{Precision} * \mathrm{Recall}}{(\mathrm{Precision}+\mathrm{Recall})}$

4.2.3Evaluation Procedure

We conduct two experiments. One usesthe patents granted in 1992 as the training data to test the patents granted in 1993, while the other utilizesthe patents granted in 1993 as the training data to test the patents granted in 1994. We randomly choose 80% of the patents of the corresponding year to be the training and testing data set respectively. Furthermore, the training-and-testing process are performed 50 times and averaged to avoid sampling bias. The results are shown and discussed in the next section.

4.3EvaluationResults

4.3.1 InitialPrediction Results before Enhancement

We apply Naïve Bayes with and without bagging as our underlying learning algorithms in the following experiments and present the evaluation statistics in **Table 12**.

Table 12 Initial Prediction Results with Different Classifiers before Enhancement

(a) Year of training/testing patents: 1992/1993				
	Accuracy	Precision	Recall	F1
Naïve Bayes	63.89%	**73.02%**	75.09%	73.98%
Bagging Naïve Bayes	**64.06%**	73.00%	**75.55%**	**74.18%**
(b) Year of training/testing patents: 1993/1994				
	Accuracy	Precision	Recall	F1
Naive Bayes	58.91%	**79.00%**	59.56%	67.84%
Bagging Naïve Bayes	**58.93%**	78.95%	**59.66%**	**67.87%**

According to the results of **Table 12**, we decide to adopt Naïve Bayes with bagging as

our underlying learning algorithmin the following evaluations.We further draw the distribution of probabilities of maintenanceof each testing patent of year 1993 under eachlearning algorithm into histograms (**Figure7**). As Figure 7 shows, most of the patents are predicted to be maintained.

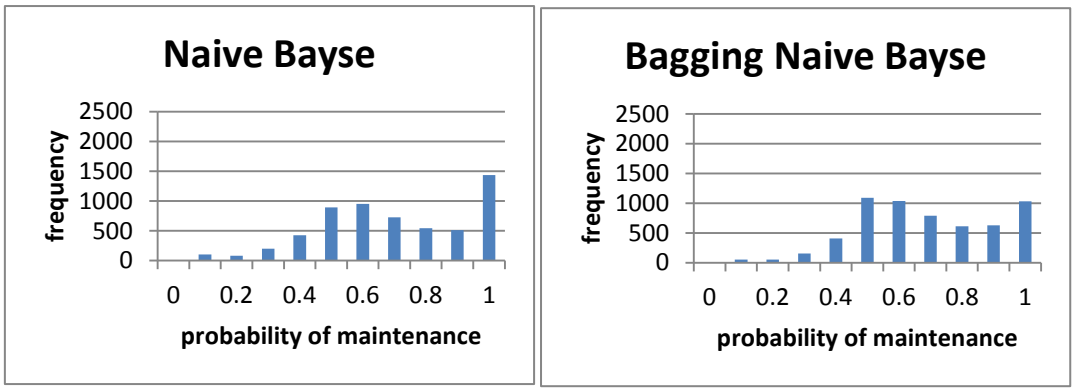


Figure 7 Distribution of Probabilities of Maintenance for Each Model

4.3.2 Effect of Weights on Classification Effectiveness Considering Company and IPC as Source Groups

In our empirical evaluation, we consider implicit similarity information (i.e., source groups) of testing patents as whether they belong to the same assignee and the same IPC classification simultaneously.Because one patent may have several IPCs, we repeatedly assign the patent to different IPC categories in computation and adopt averaged values. Besides, we consider 3-digit IPC classification here. The prediction results with different weights are illustrated in **Table 13** with Bagging Naïve Bayesas the underlying learning algorithm.

Table 13 Prediction Results with Enhancement under Different Weights Considering Company and IPC as Source Groupings

(a)Year of training/testing patents: 1992/1993							
weight	1	3	10	30	100	300	1000
Accuracy	67.76%	66.96%	66.28%	65.89%	65.81%	65.70%	65.88%
Precision	70.19%	70.60%	70.93%	70.94%	70.85%	70.82%	70.95%
Recall	92.11%	88.67%	86.09%	85.05%	85.05%	84.83%	85.05%
F1	**79.66%**	78.60%	77.77%	77.35%	77.29%	77.18%	77.34%
(b)Year of training/testing patents: 1993/1994							
weight	1	3	10	30	100	300	1000
Accuracy	66.48%	63.93%	58.24%	55.96%	54.23%	53.30%	54.79%
Precision	75.81%	75.94%	76.27%	76.78%	76.61%	76.85%	76.84%
Recall	79.50%	74.36%	62.28%	57.05%	53.83%	51.72%	54.60%
F1	**77.42%**	74.66%	68.01%	64.93%	62.77%	61.39%	63.29%

Precision improves with increasing weights while accuracy, recall, and F1 fall in both (a) and (b) experiments. The larger the weights, the stronger power we impose on the enhanced model in accordance with the majority which may lead to wrong predictions.Becausethe best F1 measurement occurs when the weight equals to 1, we set the weight to be 1 in the following experiments.

4.3.3 Comparative Evaluation Results

To show the effectiveness of the use of the enhanced classification algorithm, we adopt the original classification algorithm (i.e., without the use of the enhanced classification algorithm) as the benchmark model. Please note that Bagging Naïve Bayes is employed

as the underlying learning algorithm for both models, and the enhanced model uses the weight of 1. **Table 14** shows the comparative evaluation results.

Table 14 Comparative Evaluation Considering Company and IPC as Source Groups

(a)Year of training/testing patents: 1992/1993				
	Accuracy	Precision	Recall	F1
Benchmark model	64.06%	**73.00%**	75.55%	74.18%
Enhanced model	**67.76%**	70.19%	**92.11%**	**79.66%**
(b)Year of training/testing patents: 1993/1994				
	Accuracy	Precision	Recall	F1
Benchmark model	58.93%	**78.95%**	59.66%	67.87%
Enhanced model	**66.48%**	75.81%	**79.50%**	**77.42%**

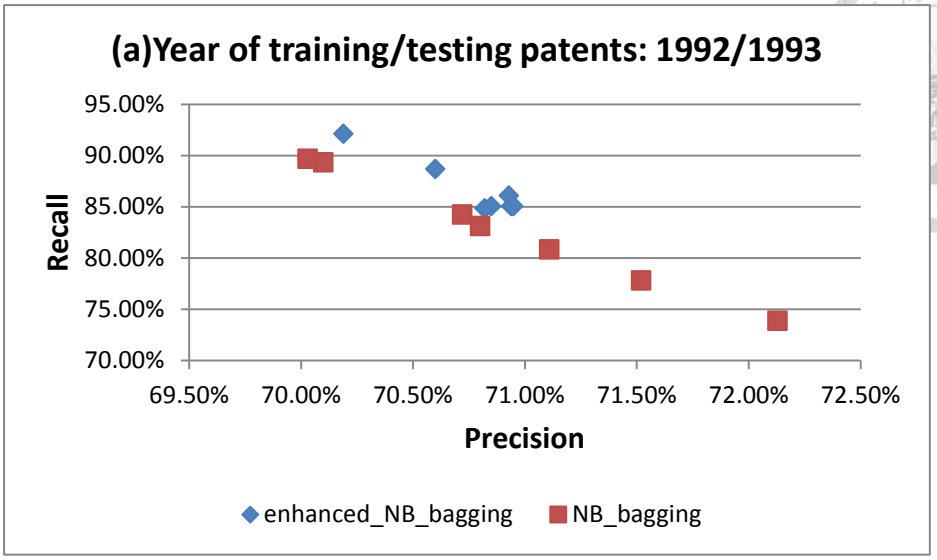
In both experiments, all measurements perform better under the enhanced model except for precision. We also conduct paired t tests to examine whether accuracy, precision, recall, and F1 of the benchmark modelare statistically significantly different from those of the enhanced model. There exist significant differences between all statistics of two models (p value < 0.001).

We can find out that our enhanced model is significantly better than the benchmark model. The results also confirm our conjecture that the implicit similarity information of testing patents can be utilized to improve the prediction of patent maintenance decisions.

To further analyze the effectiveness of our enhanced model, we depict the precision

andrecall of both models in **Figure 9**. Each point of the enhanced model represents the precision and recall with a specific weight, whereas in benchmark model each point is associated with a specific threshold to classify maintenance decisions (thresholds are 0.38, 0.4, 0.42, 0.44, 0.46, 0.48, and 0.5,respectively). That is, if the predicted probability of maintenance for a patent is larger than the threshold, the predicted decision of whether to maintain the patent will be "yes."

As **Figure 8**(a) shows, the enhanced model is superior to thebenchmark model. However, as **Figure 8**(b) demonstrates, the enhanced model with the weight greater than 3 becomes inferior to the benchmark model.



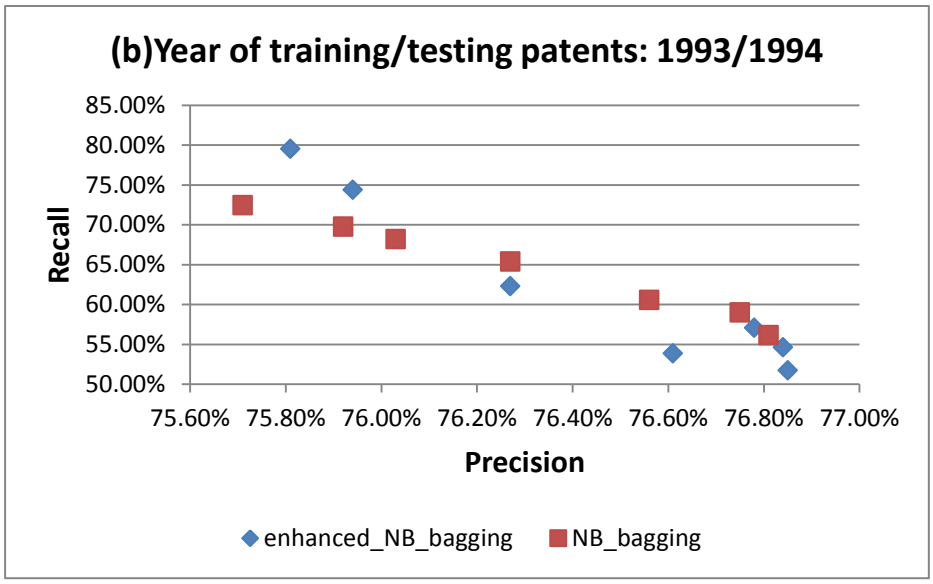


Figure 8 Precision and Recall of the Enhanced Model and the Benchmark Model

4.3.4 Precision of Top and Bottom *n*% of Testing Patents

In order to evaluate our model more precisely, we rank the testing patents according to their probability of maintenance and further examine the precision of maintenance decision "yes" and "no" respectively of top *n*% and bottom *n*% patents for each company in the two models. The average precision results from 50 rounds of

evaluationsare listed in **Table 15** and **Table 16**withBagging Naïve Bayesas the underlying learning algorithm and the weight of 1 for the enhanced model.

Table 15 Average Precision of Top and Bottom *n*%Patents with Year of Training/Testing Patents 1992/1993

		n				
		5	10	15	20	25
Average number of patents		13.86	25.23	36.11	48.38	60.55
Class Y at top *n*%	benchmark	**80.02%**	**79.28%**	**77.42%**	**76.01%**	75.49%
	enhanced	**80.02%**	**79.28%**	**77.42%**	76.00%	**75.50%**
Class N at bottom *n*%	benchmark	47.34%	42.18%	41.94%	41.01%	39.98%
	enhanced	**53.11%**	**50.59%**	**50.84%**	**48.89%**	**48.57%**

Table 16 Average Precision of Top and Bottom *n*%Patentswith Year of Training/Testing Patents 1993/1994

		n				
		5	10	15	20	25
Average number of patents		14.46	27.63	39.64	52.64	66.52
Class Y at top *n*%	benchmark	**76.86%**	79.90%	79.97%	**78.08%**	**77.53%**
	enhanced	**76.86%**	**79.91%**	**79.99%**	78.04%	77. 37%
Class N at bottom *n*%	benchmark	39.74%	**38.07%**	40.48%	39.96%	38.39%
	enhanced	**40.09%**	37.56%	**40.68%**	**40.69%**	**39.74%**

We can find that the main improvement of prediction in our enhanced model lies in the bottom testing patents. On the contrary, the performance of the top testing patents shows no notable difference among two models.That is, for those patents the initial classification model predicted to be less likely to be maintained, our enhanced model

can promote the precision of maintenance decision more obviously. Thus, companies can better apply the model to those patents with lower probabilities of being maintained.

Chapter 5 In-depth Analysis

To further investigate how different implicit similarity information we capture and how different conditions when applying the enhanced model will influence on the prediction effectiveness, we conduct several experiments in this chapter.

5.1 Experiment 1: Different Conditions of Company and IPC Source Groupings

In our previous experiment, when a patent have several IPCs, we repeatedly assign the patent to different 3-digit IPC categories. In this experiment, we consider different conditions such as 4-digit IPC classification as well as assigning patent to its first IPC classification listed on USPTO's website here with Bagging Naïve Bayes classifier and the weight of 1underthe enhanced model. The statistics are averaged values derived from 50 trials.

Table 17 Prediction Results of the Enhanced Model under Different Conditions of Company and IPC Source Groupings

(a)Year of training/testing patents: 1992/1993				
	Accuracy	Precision	Recall	F1
3-digit/multi-IPC	**67.76%**	70.19%	**92.11%**	**79.66%**
3-digit/first-IPC	67.26%	70.87%	88.69%	78.77%
4-digit/multi-IPC	67.36%	70.20%	91.01%	79.26%
4-digit/first-IPC	66.79%	**71.04%**	86.95%	78.17%
(b)Year of training/testing patents: 1993/1994				
	Accuracy	Precision	Recall	F1
3-digit/multi-IPC	**66.48%**	75.81%	**79.50%**	**77.42%**
3-digit/first-IPC	64.69%	76.62%	74.36%	75.31%
4-digit/multi-IPC	66.41%	75.76%	**79.50%**	77.38%
4-digit/first-IPC	64.20%	**77.14%**	72.45%	74.63%

The statistics show small variance among different conditions in **Table 17**. However, we can still recognize that the enhanced model with 3-digit/multi-IPC produces the best prediction results.

To investigate the difference between multiple IPCs and the first IPC, the number of groups and average number of patents in the groups under different conditions of company and IPC groups are presented in **Table 18**. Considering multiple IPCs leads to more groups and larger group size than considering only the first IPC. We believe that this is the reason why considering multiple IPCs provides the results no worse than applying only the first IPC.

Table 18 Number of Groups and Patents under Different Conditions

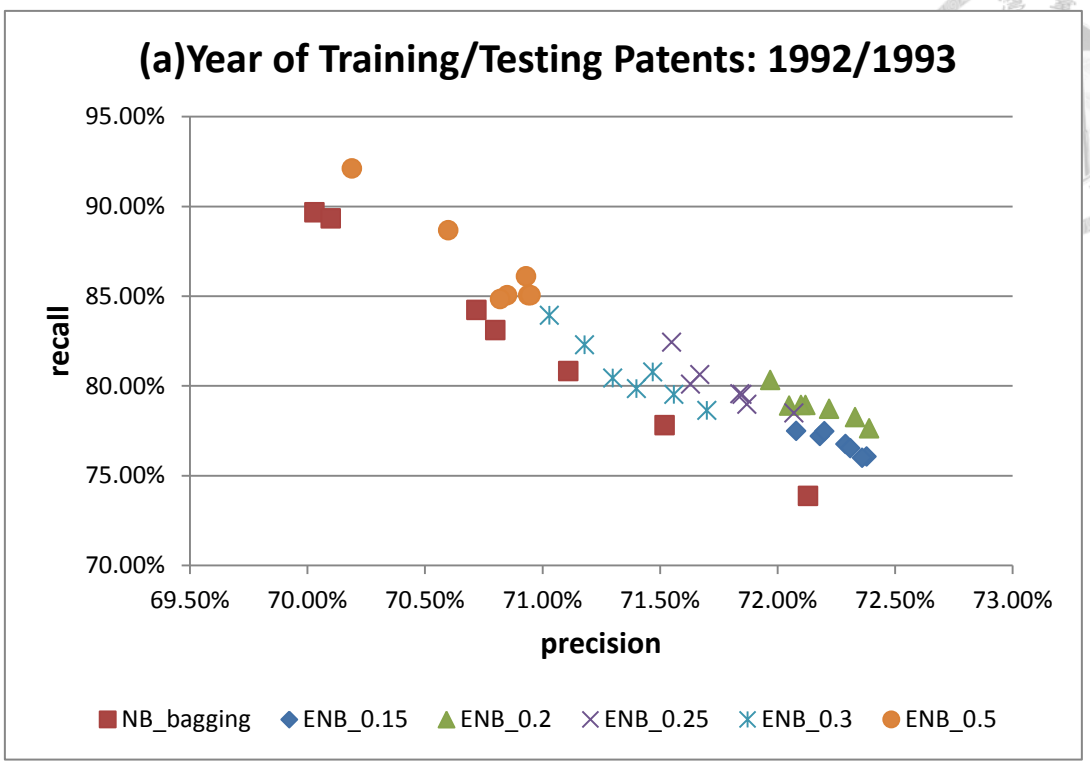
	Number of groups	**Average number of patents per group**
3-digit/multi-IPC	1036	8.85
3-digit/first-IPC	697	8.44
4-digit/multi-IPC	2345	4.57
4-digit/first-IPC	1424	4.13

5.2 Experiment 2:Extension of the Enhanced Model

The larger weight does not provide better prediction results. As a consequence, we consider applying theenhanced model only under some circumstances. For each company and IPC source group, if the maintenance rate in previous year is relative large or small, the tendency to maintain or abandon patents may be more consistent and obvious so that the enhanced model should be utilized.

In this section, we merely apply the enhanced model to the testing patents in the company and IPC source groups whose maintenance rate in previous year is either larger than 1-α oris smaller than α (α lies between 0 and 0.5), leaving those whose probabilityfalls between1-α and α unadjusted (i.e., followingthe initial predictions). Each data point of *ENB_α* represents the precision and recall of the prediction with Bagging Naïve Bayes classifier and the weight of 1 underthe enhanced model. Furthermore, the classification processes are performed 50 times and averaged to avoid

sampling bias.When α equals 0.5, it means all testing patents would apply the enhanced model. *NB_bagging* stands for the benchmark models with different thresholds to classify maintenance decision showed previously in **Figure 9**(a). For those company and IPC source groups which cannot find any training data in previous year as measurement, we apply the enhanced model still and the comparative results are shown in **Figure 9(a)** and **Figure 9(b)**.Our enhance model outperforms the benchmark models in each selected αin **Figure 9(a)** and **9(b)**. As αincreases, precision will be promoted as well while recall decreases on the contrary.



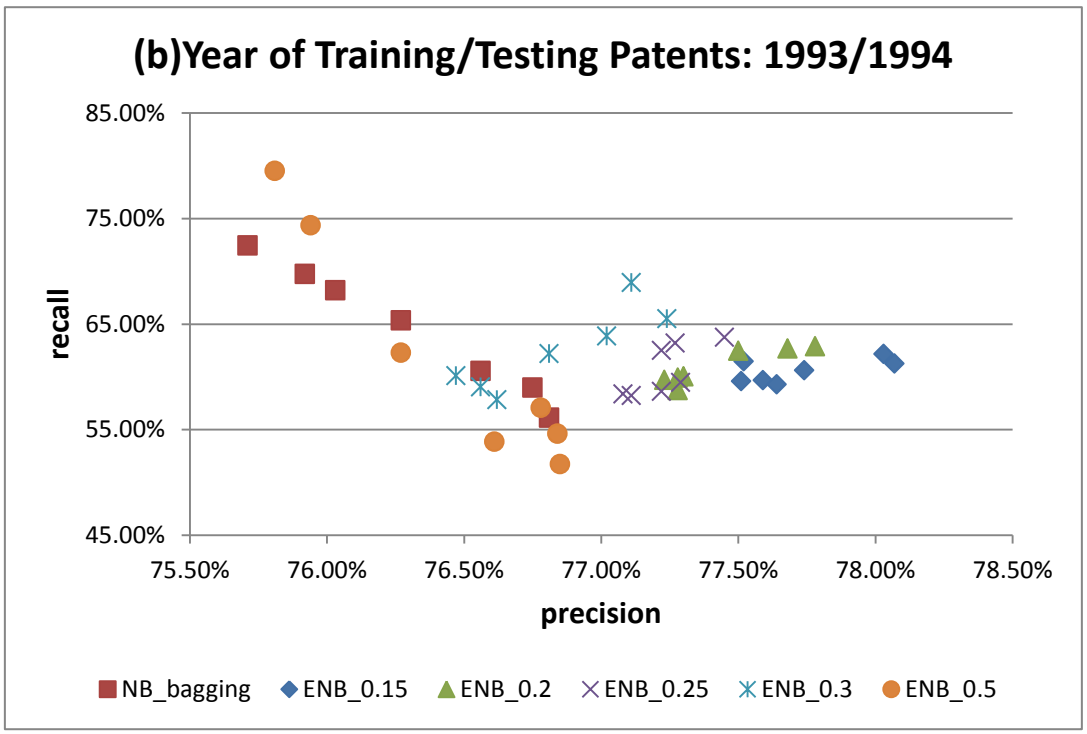


Figure 9 Comparative Evaluation under theEnhanced Model with Different α

Chapter 6 Conclusion and Future Directions

6.1 Conclusion

In this research, we apply23 factors grouped into 7 dimensions including*business strategy, commercialization, patent family, patent defensiveness, technology type, future R&D, and industries*to build the classification model to predict the maintenance decisions ofpatents in the third stage. These factors capture the prominent characteristics of patents and provide insight into determining the maintenance decisions.

We build an enhanced model in predicting maintenance decisions using those factors with the data mining approach. The enhanced algorithm considers the implicit similarity grouping of patents to be predicted, namely company and IPC, so that the model can reach better prediction performance.

Our empirical results suggest that our proposed enhanced model outperforms the benchmark model without applying the enhanced algorithm.Considering every 3-digit IPC the patent belongs to would contribute to better performance. These experiments confirm the effectiveness of our proposed enhanced model to support the third stage patent maintenance decisions.

6.2 Future Directions

First, we retrieve patents regarding to its assignee. However, a company may have several legal titles so that there should exist some patents being excluded in our empirical research. Thus, the completeness of our evaluation can be improved by developing a list of alternative names for each company to better capture target patents.

Second, although we employ variables suggested in the literature to reflect the characteristics of the patents to be maintained, there may be some other essential factors that are not included in our research such as the number of transferring of assignees. For some dimensions, patent use, for example,is hard to be capturedby indicators derived from a patent document. As a result, developing additional variables to reflect more patent maintenance factors is worth further investigation.

Third, the proposed enhanced model intends to improve the prediction accuracy through implicit similarity of testing patents. So the performance of the model highly depends on what similarity we group testing patents into. In our research, we choose assignee and IPC codes. Other grouping factors such as inventors can be applied in further analysis. However, no matter what grouping method we utilize, the number of patents in each group should not be insufficient to effectively compute the corresponding probabilities used in the enhanced model.

Finally, there are still some other enhanced algorithms which can be utilized for

improving the classification results. In consequence, trying different algorithms would probably contribute to divergent outcomes and deserves more research attentions.

References

Agrawal, R., & Srikant, R. (2001). On integtaing catalogs. *Proceedings of the 10th International Conference on World Wide Web*, 603–612.

Albert, M. B., & Avery, D. (1990). Direct validation of citation counts as indicators of industrially important patents. *Research Policy*, *20*(3), 251–259.

Bessen, J. (2008). The value of U.S. patents by owner and patent characteristics. *Research Policy*, *37*(5), 932–945.

Breiman, L. (1996). Bagging predictors. *Machine Learning*, *140*(24), 123–140.

Chang, C.T. (2010). Supporting patent maintenance decision: A data mining approach. Unpublished Master Thesis, Institute of Service Science, National Tsing Hua University.

Choksi, J. (1999). The benefits and costs of patent protection. *IEEE Canadian Review - Summer*, 25–26.

Davenport, T. H., DeLong, D. W., & Beers, M. C. (1998). Successful knowledge management projects.*Sloan Management Review*, *39*(2), 43–57.

Fall, C. J., Törcsvári, A., Benzineb, K., & Karetka, G. (2003). Automated categorization in the international patent classification. *ACM SIGIR Forum*, *37*(1), 10–25.

Griliches, Z. (1998). Patent statistics as economic indicators: A survey. Chapter 13 in *R&D and Productivity: The Econometric Evidence*, Z. Griliches (ed.), University of Chicago Press,287–343.

Hall, B. H., Jaffe, A. B., &Trajtenberg, M. (2001). The NBER patent citations data file: Lessons, insights and methodological tools. NBER Working Paper SeriesNo. 8498.

Harhoff, D., Narin, F., Scherer, F. M., & Vopel, K. (1999). Citation frequency and the value of patented inventions. *The Review of Economics and Statistics*, *81*(3), 511–515.

Jin, X., Spangler, S., Chen, Y., Cai, K., Ma, R., Zhang, L., Wu, X., & Han, J. (2011). Patent maintenance recommendation with patent information network model. *Proceedings of 2011 IEEE 11th International Conference on Data Mining*, 280–289.

Jun, S., & Uhm, D. (2013). A predictive model for patent registration time using survival analysis. *Applied Mathematics & Information Sciences*, *7*(5), 1819–1823.

Lanjouw, J. O., & Schankerman, M. (2004). Patent quality and research productivity: measuring innovation with multiple indicators. *The Economic Journal*, *114*(495), 441–465.

Li, X., Chen, H., Zhang, Z., Li, J., & Nunamaker, J. F. (2009). Managing knowledge in light of its evolution process: An empirical study on citation network-based patent classification. *Journal of Management Information Systems*, *26*(1), 129–154.

Liu, K., Arthurs, J., Cullen, J., & Alexander, R. (2008). Internal sequential innovations: How does interrelatedness affect patent renewal?,*Research Policy*, *37*(5), 946–953.

Nikzad, R. (2011). Survival analysis of patents in canada. *The Journal of World Intellectual Property*, *14*(5), 368–382.

Oh, S., Lei, Z., Mitra, P., & Yen, J. (2012). Evaluating and ranking patents using weighted citations. *Proceedings of the 12th ACM/IEEE-CS Joint Conference on Digital Libraries*, 281–284.

Pakes, A. (1986). Patents as options: Some estimates of the value of holding european patent stocks. *Econometrica*, *54*(4), 755–784.

Schankerman, M., & Pakes, A. (1985). Estimates of the value of patent rights in European countries during the post-1950 period. NBER Working Paper Series No.1650.

Shih, M.J., Liu, D.R., & Hsu, M.L. (2010). Discovering competitive intelligence by mining changes in patent trends. *Expert Systems with Applications*, *37*(4), 2882–2890.

Teece, D. J. (1986). Profiting from technological innovation : Implications for integration, collaboration, licensing and public policy. *Research Policy*, *15*(6), 285–305.

Jones, T., Norris, M., & Solomon, I. (2002). Strategies for maximizing value from intellectual capital in a technology-driven business. *The Licensing Journal*, *22*(6), 1–6.

Thomas, P. (1999). The effect of technological impact upon patent renewal decisions. *Technology Analysis & Strategic Management*, *11*(2), 181–198.

Trappey, A. J. C., Hsu, F.C., Trappey, C. V., & Lin, C.I. (2006). Development of a patent document classification and search platform using a back-propagation network. *Expert Systems with Applications*, *31*(4), 755–765.

van Zeebroeck, N. (2009). From patent renewals to applications survival: do portfolio management strategies play a role in patent length?, Working Papers CEB, ULB -- Universite Libre de Bruxelles, No. 09-028.RS.

Wang, M., Lo, H., Liao, Y., & Lin, P. (2012). Determinants of patent renewal decisions by patent Indicators and social network analysis : The case of the biotech industry in Taiwan and Korea.*Proceedings of Technology Management for Emerging Technologies (PICMET)*, 1060–1065

Yoon, B., & Park, Y. (2004). A text-mining-based patent network: Analytical tool for high-technology trend. *The Journal of High Technology Management Research*, *15*(1), 37–50.

Appendix

Exhibit 1 Reference Source of Variables in Our Study

Dimensions	Variables	Reference
Business Strategy	Maintenance compatibility of tech scope (IPC)	(Chang, 2010)
	Abandonment compatibility of tech scope (IPC)	(Chang, 2010)
Commercialization	Tech scope (IPC)	(Jin et al., 2011; Nikzad, 2011; Oh et al., 2012)
	Tech scope (USPC)	(Jin et al., 2011)
Future R&D	Internal sequential innovation	(Liu et al., 2008)
	Forward self-citation rate	(Albert & Avery, 1990; van Zeebroeck, 2009)
	Backward self-citation rate	(Harhoff et al., 1999; Thomas, 1999)
Patent Defensiveness	Number of claims	(Bessen, 2008; Jin et al., 2011; van Zeebroeck, 2009)
	Number ofindependent claims	(Jin et al., 2011; van Zeebroeck, 2009)
	Approval time	(Chang, 2010)
Patent Family	Number ofbackward citations_US	(Harhoff et al., 1999; Thomas, 1999)
	Number ofbackward citations	(Harhoff et al., 1999; Jin et al., 2011; Thomas, 1999)
	Science links	(Chang, 2010; Harhoff et al., 1999)
	Number offorward citations	(Albert & Avery, 1990; Harhoff et al., 1999; Thomas, 1999; van Zeebroeck, 2009)
	Family size (countries)	(Harhoff et al., 1999; Lanjouw & Schankerman, 2004; van Zeebroeck, 2009)
	Family size (applications)	(Harhoff et al., 1999; Lanjouw & Schankerman, 2004; van Zeebroeck, 2009)
	Major market index	(Chang, 2010; Wang et al., 2012)

Technology Type	Generality index (IPC)	(Chang, 2010; Hall et al., 2001; Jin et al., 2011; Shih et al., 2010; Wang et al., 2012)
	Originality index (IPC)	(Chang, 2010; Hall et al., 2001; Jin et al., 2011; Wang et al., 2012)
	Cited half-life (median)	(Chang, 2010)
	Maintenance of backward citations	(Chang, 2010)
	# of cross state cooperation	(Chang, 2010)
Industry	Industry	(Chang, 2010; Shih et al., 2010)

Exhibit 2 Industries and Corresponoing Companies in Our Study

Industry	Company	Global 500 rank
Aerospace and Defense	Boeing	76
	EADS	109
	United Technologies	120
	Lockheed Martin	135
	Northrop Grumman	166
	Honeywell Intl.	206
	Raytheon	292
	General Dynamics	308
	BAE Systems	359
	Bombardier	379
Chemicals	BASF	91
	Dow Chemical	104
	Bayer	124
	DuPont	188
	Mitsubishi Chemical	288
	Sabic	331
	Akzo Nobel	380
	Hanwha	393
	Asahi Kasei	483
Electronics, Electrical Equipment	Siemens	21
	Hitachi	23
	Matsushita Electric Industrial	25
	Samsung Electronics	39
	Sony	47
	Toshiba	72
	Tyco International	103
	LG Electronics	115
	Royal Philips Electronics	116
	Mitsubishi Electric	156
Motor Vehicles & Parts	General Motors	5

	DaimlerChrysler	6
	Toyota Motor	7
	Ford Motor	8
	Volkswagen	15
	Honda Motor	27
	Nissan Motor	29
	Peugeot	41
	Fiat	57
	BMW	71
Network and Other Communications Equipment	Nokia	130
	Motorola	138
	Cisco Systems	254
	L.M. Ericsson	338
	Alcatel	399
Pharmaceuticals	Pfizer	75
	Johnson & Johnson	88
	GlaxoSmithKline	122
	Novartis	186
	Roche Group	209
	Merck	239
	Bristol-Myers Squibb	258
	AstraZeneca	267
	Abbott Laboratories	285
	Sanofi-Aventis	321
Semiconductors and Other Electronic Components	Intel	141
	Flextronics International	375
	Onex	457
	Solectron	478
	Texas Instruments	491
Telecommunications	Nippon Telegraph & Telephone	18
	Deutsche Telekom	37
	Verizon Communications	38

	Vodafone	53
	France Télécom	63
	SBC Communications	102
	Telecom Italia	111
	Telefónica	114
	BT	140
	AT&T	162